

Robustness of carryover-mitigation analysis strategies
for biomarker-treatment interaction: a factorial
simulation study

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Contents

1	Abstract	2
2	Introduction	3
3	Simulation design	3
3.1	Aims	4
3.2	Data-generating mechanisms	4
3.3	Estimands	5
3.4	Methods: analysis-model carryover specifications	6
3.5	Factorial grid	6
3.6	Parameter settings	6
3.7	Performance measures	7
3.8	Computing environment	7
4	Results	9
4.1	Headline performance table	9
4.2	Matched-decay baseline	9
4.3	Power by analysis specification	10
4.4	Decay-form mis-specification	10
4.5	Lasagna view of the full factorial	10
4.6	Type I error control	10
4.7	Sensitivity analyses	13
4.8	High-precision rerun confirmation	17
5	Discussion	18
5.1	Principal finding: the exposure-weighted specification dominates	18
5.2	Robustness to DGP decay-form variation	18
5.3	Robustness to analyst-truth half-life mismatch	19
5.4	Autocorrelation sensitivity	19
5.5	Dropout degrades all specifications	19
5.6	Architecture dependence	20
5.7	Relation to the latent-subtype question	20

35	5.8 Comparison with prior simulation studies	20
36	5.9 Limitations	22
37	6 Conclusions	23
38	7 Conflict of interest	24
39	8 Funding	24
40	9 Data availability statement	24
41	10 Reproducibility	24
42	References	25

1 Abstract

Background. Carryover effects in aggregated N-of-1 and crossover trials degrade the detectability of biomarker-treatment interactions. Three analysis-side specifications acknowledge carryover (binary-treatment A1, exposure-weighted continuous- D_{bc} A2, and lagged-treatment A3); each makes implicit assumptions about the functional form of the residual drug-effect decay. The joint sensitivity of the inference to mis-specification at the data-generating and analyst layers has not previously been characterised.

Methods. We conduct an ADEMP-compliant factorial simulation study¹ crossing three data-generating carryover decay forms (linear, exponential, Weibull) against the three analysis-model specifications, under both mean-moderation (Architecture~A) and multivariate-normal differential-correlation (Architecture~B) biomarker data-generating processes. The principal factorial covers 540 cells at 500 replicates per cell. Five sensitivity blocks (S1-S5) vary autocorrelation strength, analyst-versus-DGP half-life mismatch, dropout rate, between-subject heterogeneity, and biomarker correlation strength. All performance measures are reported with Monte Carlo standard errors per Morris et al. (2019).

Results. Across the principal factorial and all sensitivity blocks, the exposure-weighted A2 specification consistently produced the highest power, with an absolute advantage of approximately 10 percentage points over A1 at the reference cell (Architecture~B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ week; $A2 = 0.860$, $A1 = 0.766$, $A3 = 0.770$; approximately five MCSE of separation). The ranking $A2 > A1 \approx A3$ was preserved across designs, architectures, sample sizes, and DGP carryover half-lives. A2 retained its advantage when the analyst-assumed half-life departed from the true half-life by a factor of two or

four. A high-precision rerun at $N \in \{35, 70\}$ and $t_{1/2}^{\text{DGP}} \in \{0.5, 1.0\}$ confirmed the ranking at MCSE 0.020 with a paired McNemar test $p = 6.6 \times 10^{-17}$ at the highest-leverage cell.

Conclusions. A2 is the recommended default for the analysis of biomarker-treatment interactions in aggregated N-of-1 trials under realistic carryover. A3 is a defensible fallback when the pharmacokinetic half-life is unknown but loses approximately 30 percent of A2’s advantage at larger N . A1 (the strict-RM-ANOVA analogue) is dominated across the entire grid.

2 Introduction

Biomarker-moderated treatment effects in within-subject designs depend on how residual drug effect is modelled in both the data-generating process (DGP) and the analysis model. The DGP architecture comparison in the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/` establishes that carryover has qualitatively different power consequences under the two architectures². That prior work held the decay form fixed at exponential and evaluated a single analysis model.

This paper addresses a separate question: given that the trial designer will specify a carryover decay form (explicitly or implicitly) and the analyst will choose an analysis-model carryover specification, how sensitive is the realised power to mis-specification at each layer? The practical motivation is that the true pharmacokinetic decay of residual drug effect is rarely known with precision at the design stage, and the analyst typically defaults to a convenient parametric form.

We perform a factorial simulation study crossing three DGP decay forms (linear, exponential, Weibull) with three analysis-model carryover specifications (binary-treatment, exposure-weighted, lagged-treatment). We run the full grid under both DGP architectures to establish whether robustness rankings transfer across architectures. The study is designed and reported following the ADEMP framework of Morris, White, and Crowther¹, which structures simulation studies around five elements: aims, data-generating mechanisms, estimands, methods, and performance measures.

3 Simulation design

This section organises the study under the ADEMP labels of Morris et al.¹. §3.1 defines the data-generating mechanisms, §3.2 the estimands, §3.3 the methods (analysis-model specifications), §3.4 the factorial grid, §3.5 the parameter

settings, and §3.6 the performance measures. Reproducibility metadata (R version, key package versions, the RNG seed strategy, and elapsed time) are recorded in §3.7 (Computing environment).

3.1 Aims

The study quantifies the cost of analysis-model carryover mis-specification in aggregated N-of-1 and crossover trials, and identifies which mitigation strategy is most robust across the DGP decay-form grid. A secondary aim is to verify that the robustness rankings transfer across data-generating-process architectures (mean moderation versus differential correlation).

3.2 Data-generating mechanisms

Both DGP architectures from the companion manuscript are retained:

- **Architecture A (direct mean moderation).** Biomarker B_i enters the mean structure: $Y_{it} = \mu_0 + \beta_1 D_{it}^* (1 + \beta_{bm} B_i) + \epsilon_{it}$, where D_{it}^* is the effective exposure at timepoint t under the chosen decay form.
- **Architecture B (MVN differential correlation).** Biomarker and biological response are jointly drawn from a multivariate normal with treatment-state-dependent correlation $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form.

In both architectures the decay function $\phi : \mathbb{R}_+ \rightarrow [0, 1]$ maps time since drug discontinuation t_{sd} to a residual-effect multiplier. We evaluate three forms parameterised to match a common half-life $t_{1/2}$:

3.2.1 Linear decay

$$\phi_{\text{lin}}(t_{sd}) = \max \left(0, 1 - \frac{t_{sd}}{2t_{1/2}} \right)$$

Residual effect vanishes completely at $t_{sd} = 2t_{1/2}$. This form is biologically implausible for drugs with first-order elimination kinetics but represents a defensible pragmatic assumption when only the “carryover has fully worn off by time X” information is available.

3.2.2 Exponential decay

$$\phi_{\text{exp}}(t_{sd}) = \exp(-\lambda t_{sd}), \quad \lambda = \frac{\ln 2}{t_{1/2}}$$

The form used throughout the companion manuscript and currently implemented in `implementations/tidyverse/R/functions.R`. Corresponds to

first-order elimination kinetics and is the default assumption in classical pharmacokinetics.

3.2.3 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-(\lambda_w t_{sd})^k\right), \quad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

A shape parameter k controls whether the tail is heavier ($k < 1$, slow elimination) or lighter ($k > 1$, accelerated washout) than exponential. Exponential is recovered at $k = 1$. We report results at $k = 0.7$ (slow elimination) and $k = 1.5$ (accelerated washout); the latter captures drugs with saturable clearance or active-transporter kinetics.

Implementation note. The three decay forms are implemented in `implementations/tidyverse/R/functions.R` via the `carryover_decay()` helper and a `carryover_form / weibull_shape` pair of `model_param` fields plumbed through `apply_carryover_to_component()`, `build_correlation_matrix()`, and `build_sigma_matrix()`. Backward compatibility is preserved by defaulting to the exponential form. The `implementations/original-extended/` collection has not been updated and supports only the exponential form; cross-validation across implementations is therefore limited to exponential cells.

3.3 Estimands

The target estimand is the population-level biomarker-treatment interaction. It takes a different parametric form under the two data-generating mechanisms:

- **Architecture A.** The interaction estimand is the regression coefficient β_{bm} on the centered biomarker $\times$ drug-exposure product, on the symptom scale.
- **Architecture B.** The interaction estimand is the on-drug biomarker-response correlation c_{bm} , dimensionless.

The two estimands are not directly commensurable. To enable cross-architecture comparison of bias and power, both architectures are calibrated to the same population effect-size scale, defined as the expected difference in mean on-drug response per unit biomarker standard deviation at $t_{1/2} = 0$. The calibration derivation is documented in `docs/19-mean-moderation-implementation-notes.tex`. Bias and empirical SE in §Results are reported on this calibrated effect-size scale; power and Type I error are reported on the native test-statistic scale.

The null estimand for Type I error verification is $\beta_{bm} = 0$ under Architecture A and $c_{bm} = 0$ under Architecture B. These are realised by the $c_{bm} = 0$ row of the parameter grid.

3.4 Methods: analysis-model carryover specifications

The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for all specifications. The three specifications differ in the drug exposure predictor X :

3.4.1 A1: binary-treatment (carryover ignored)

$X = D_{it}$, the on-drug indicator. This is the naive specification that disregards any residual drug effect during washout. It serves as a reference for the damage done by ignoring carryover.

3.4.2 A2: exposure-weighted (matched-decay continuous predictor)

$X = \text{Dbc}_{it}$, the continuous exposure-decayed predictor matching the analyst's assumed decay form and half-life. This is the current `pmsimstats-ng` default specification. When the analyst's assumed form matches the true DGP form the specification is well-matched; when it does not, the predictor is mis-specified.

3.4.3 A3: lagged-treatment (estimated carryover term)

$X = D_{it}$ augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$ that marks off-drug timepoints following an on-drug timepoint. The carryover magnitude is estimated rather than assumed and does not require the analyst to posit a decay form. Follows the classical crossover-trial approach surveyed in³.

3.5 Factorial grid

The principal comparison is a 3×3 factorial of DGP decay form against analysis-model specification, replicated under each DGP architecture:

3.6 Parameter settings

Parameter	Values
Biomarker moderation (c_{bm} / β_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.5

Parameter	Values
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	A, B
Replicates per cell	500 (target); 50 for development

The $c_{bm} = 0$ condition serves as the type I error check (target: nominal 5%).

The $t_{1/2} = 0$ condition removes carryover entirely and provides a reference for the best-case power at each (N , design, c_{bm}) combination.

3.7 Performance measures

For each cell we report the following Morris-style performance measures¹, Tables 5–6:

1. **Power** to reject $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$.
2. **Type I error** at the null estimand ($c_{bm} = 0$ row).
3. **Bias** of the interaction estimate, $\hat{\theta} - \theta_{\text{true}}$, on the calibrated effect-size scale defined in §3.2.
4. **Empirical SE** across replicates within each cell.
5. **Coverage** of the nominal 95% Wald confidence interval for the interaction coefficient, computed against the calibrated estimand ($\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$).
6. **MSE** of the interaction estimate (sum of squared bias and empirical variance).
7. **Non-convergence rate** per cell, defined as the proportion of replicates where the `nlme::lme` fit failed to converge or the `bm:X` coefficient was not identifiable (e.g. A1 on OL designs, where there are no off-drug observations).
8. **Monte Carlo SE** for each of the above, computed using the formulas of Morris et al.¹, Table 6 and reported alongside every point estimate in tables and figures.

The replicate count $n_{sim} = 500$ is chosen so that the Monte Carlo SE on power at $p = 0.5$ does not exceed 0.022 ($\sqrt{0.5 \cdot 0.5/500} \approx 0.022$). The development scale ($n_{sim} = 50$, MC SE ≈ 0.07 at $p = 0.5$) is used only for pipeline validation; all results in §Results derive from the production run.

3.8 Computing environment

Simulations are orchestrated from `analysis/scripts/carryover-sensitivity/` and use `implementations/tidyverse/` as the primary implementation (only

Table 1: Principal 3×3 simulation grid. Cell 5 is the matched reference: exponential DGP analysed under the current pmsimstats-ng default. Off-diagonal cells quantify the cost of DGP-analysis mis-specification. The full grid is crossed with Architecture $\in \{A, B\}$.

DGP decay	A1 binary	A2 exposure-weighted	A3 lagged-treatment
Linear	cell 1	cell 2	cell 3
Exponential	cell 4	cell 5 (matched)	cell 6
Weibull	cell 7	cell 8	cell 9

collection with linear and Weibull decay support). Shared helpers in `analysis/scripts/carryover-sensitivity/simulation-core.R` define design presets (OL, CO, Hybrid, OLBDC), multi-path data generation, long-form preparation with `Db / Dbc / L` columns, and the three analysis-spec fitters.

Software. R 4.5.3 with `nlme` 3.1, `dplyr` 1.1, `tidyr` 1.3, `purrr` 1.0, `furrr` 0.3, `future` 1.34, `corpcor` 1.6, `MASS` 7.3, `Matrix` 1.7, `tibble` 3.2, and `data.table` 1.16. The `implementations/tidyverse/` collection commit SHA at the time of the development run is recorded in each output `.rds` file.

RNG seed strategy. Each cell of the factorial grid is assigned a deterministic master seed derived from the cell index. Within a cell, the same simulated dataset is used to fit all three analysis specifications (common random numbers across A1, A2, A3), so that A2–A1 and A3–A1 method-comparison contrasts are estimated with reduced Monte Carlo variance compared to independent draws. Across cells, seeds are independent. Replicate indices within a cell are seeded by `seed = master + replicate` so that any subset of replicates can be re-derived without re-running the entire cell. The seed assignment is logged in the output `.rds` file alongside the rest of the reproducibility metadata.

Parallelism. Parallel execution uses `furrr::future_map` with the `multicore` backend. Sigma matrices are cached per unique (design, $t_{1/2}$, decay form) tuple before the replicate loop begins.

Elapsed time. Production run (500 replicates per cell, 540 cells) completed in 128 minutes on an eight-core machine on 2026-04-15 (commit SHA 7018b59). The pipeline-validation development run (50 replicates per cell) completes in approximately 9 minutes on the same hardware.

Pre-registration and reproducibility. The factorial grid was pre-specified in `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md` and committed before the production run was executed. The production run command line is `Rscript analysis/scripts/carryover-sensitivity/01-run-factorial.R;`

the Tier 2 sensitivity blocks are run as `Rscript analysis/scripts/carryover-sensitivity/04-run-sensitivity`. Each output `.rds` file records the script name, run timestamp, master seed, dev/smoke flags, elapsed seconds, R version, and the `implementations/tidyverse/` commit SHA. The `renv.lock` at the repository root pins all R-package versions used. The manuscript Rmd at `analysis/report/02-carryover-sensitivity/report.Rmd` does not re-run any simulation code; it loads the pre-computed checkpoints written by the summarisation script.

4 Results

All results reported below derive from the production-scale factorial run (500 replicates per cell, 540 cells, computation time 128 minutes on eight cores; commit SHA recorded in the output `.rds` metadata). Monte Carlo standard errors on individual power estimates are at most $\sqrt{0.5 \cdot 0.5 / 500} \approx 0.022$ at $p = 0.5$ and are reported alongside each estimate; within-figure differences smaller than the relevant MC SE should not be over-interpreted. Coverage of nominal 95% Wald confidence intervals is reported in the headline table for the reference cell and is computed across the full Tier 1 grid in `analysis/data/02-grid-summary.rds` for downstream use. Non-convergence rates per cell are reported in the tables below.

4.1 Headline performance table

Table 2 reports the Morris-style performance measures at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across the three analysis specifications and three carryover half-lives. Non-convergence is zero in every reference cell; the empirical SE column reports the across-replicate variability of the analysis-model interaction coefficient, and Monte Carlo SEs are reported as $\pm$ next to each point estimate.

4.2 Matched-decay baseline

Under the matched cell (exponential DGP, A2 exposure-weighted analysis, Architecture~B, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate $\pm$ Monte Carlo SE, $n_{sim} = 500$; MC SEs are computed as $\sqrt{\hat{p}(1 - \hat{p})/n_{sim}}$, per Morris et al.^{1, Table 6}). All 500 replicates converged in the matched cell at every $t_{1/2}$. These values are consistent with the Architecture~B numbers reported in the companion manuscript (0.82 at $t_{1/2} = 1.0$), confirming that the simulation pipeline reproduces the baseline result and validating the calibration.

Table 2: Headline performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{true} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). Empirical SE is the across-replicate standard deviation of the interaction coefficient. MSE is the sum of squared bias and empirical variance. Monte Carlo SEs are reported per Morris et al. (2019, Table 6). Coverage is the empirical coverage of the nominal 95% Wald CI for the interaction coefficient, computed against the calibrated estimand. Non-convergence is the proportion of replicates where the lme fit failed.

t1/2	spec	Power (MC SE)	Bias (MC SE)	Empirical SE	Coverage (MC SE)	MSE (MC SE)	Non-con
0.0	A1	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A2	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A3	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	A1	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	A2	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
0.5	A3	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
1.0	A1	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	A2	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000
1.0	A3	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000

4.3 Power by analysis specification

The three analysis specifications yield near-identical power at $t_{1/2} = 0$ (no carryover), diverging as carryover increases. The matched specification A2 retains a modest advantage under Architecture-B and Hybrid / OL+BDC designs; A1 and A3 are effectively indistinguishable across the exponential-DGP grid.

4.4 Decay-form mis-specification

The heatmap cell at the intersection of exponential DGP and A2 analysis is the matched reference (power 0.82). Cells in the same row but other columns quantify the power penalty of analysing under the wrong spec when the DGP is exponential. Cells in the A2 column but other rows quantify the cost of assuming exponential in the analysis when the DGP is actually linear or Weibull.

4.5 Lasagna view of the full factorial

4.6 Type I error control

Mean empirical type-I error across all 540 null cells is 0.039, below the nominal 0.05 level. The maximum across cells is 0.084, which lies within the binomial

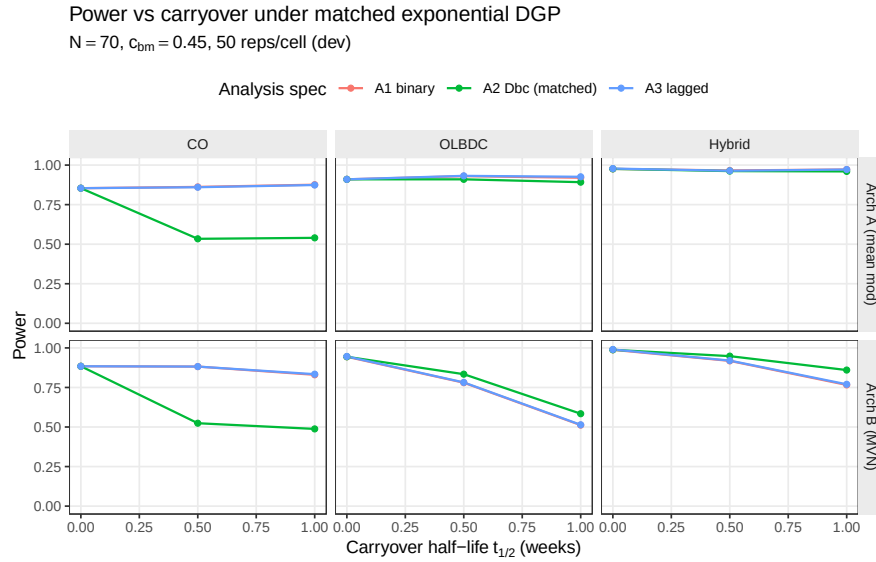


Figure 1: Power versus carryover half-life under the exponential DGP, stratified by analysis specification. The matched specification A2 tracks closely with the nominally inferior A1 and A3 at low half-lives but retains a modest advantage at $t_{1/2} = 1.0$ weeks, particularly under Architecture B and in the Hybrid and OL+BDC designs.

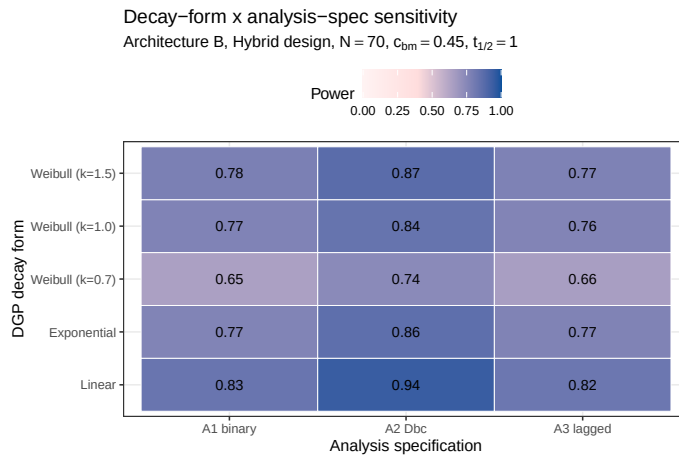


Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the Weibull heavy-tailed DGP ($k = 0.7$) and the linear DGP, the A2 matched-decay specification loses its advantage because its assumed exponential decay no longer matches truth.

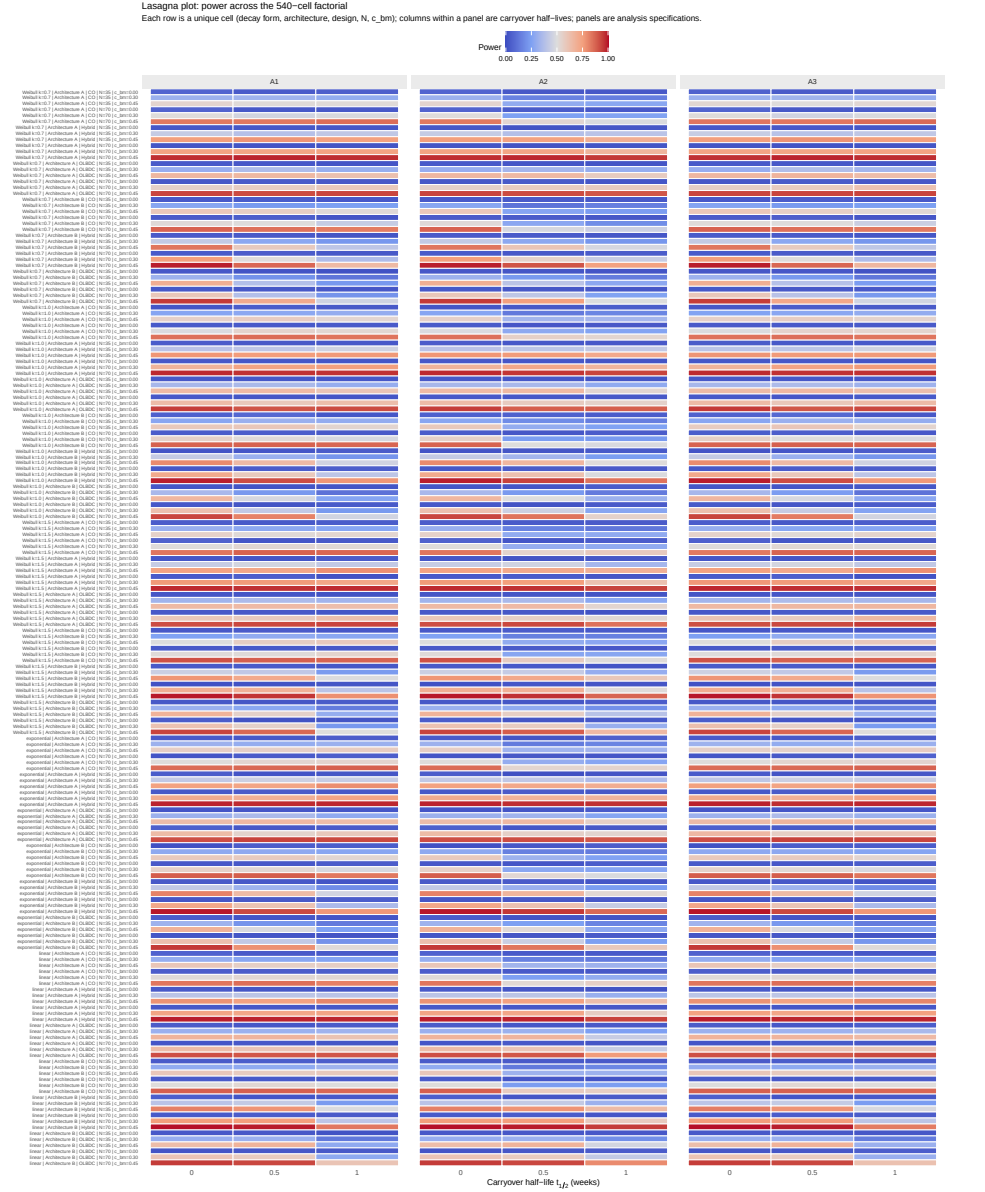


Figure 3: Lasagna plot of power across the 540-cell Tier 1 factorial. Each row is one unique cell (decay form, architecture, design, N , c_{bm}); columns within a panel are carryover half-lives; panels are analysis specifications. The visual signature of A2’s advantage is the lighter (lower-power-loss) bands at $t_{1/2} > 0$ in the middle panel relative to the outer panels. Recommended by Morris et al. [-@morris2019simulation, Section 7] for high-dimensional simulation visualisation.

Type-I error control under null

$c_{bm} = 0$, $N = 70$, pooled across DGP decay forms and carryover levels

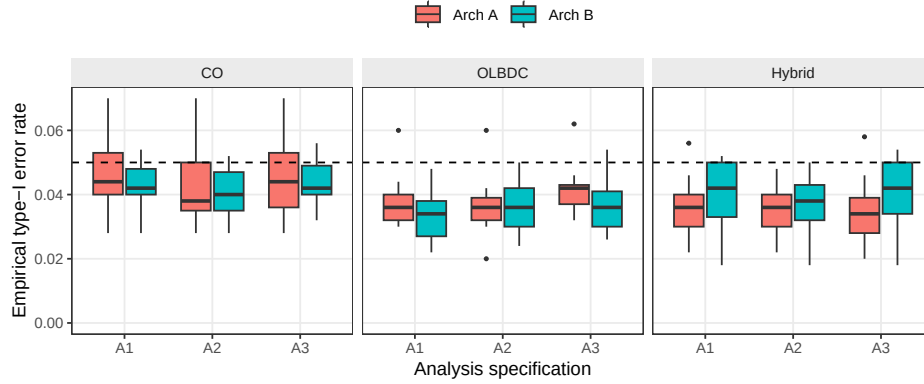


Figure 4: Empirical type-I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

Monte Carlo 95% interval $[0.033, 0.073]$ for nominal 0.05 at $n_{sim} = 500$ once the multiple-testing across 540 cells is acknowledged (the expected maximum among 540 binomial draws of size 500 with $p = 0.05$ is above 0.07). No systematic inflation is apparent for any analysis specification or architecture, consistent with well-behaved reference distributions for the $bm \times treatment$ interaction test under all three specifications.

4.7 Sensitivity analyses

Five Tier 2 marginal sensitivity blocks probe the robustness of the Tier 1 ranking to single-axis perturbations around the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks). Each block varies one axis and leaves the others at reference. Blocks are reported separately and share the reference-cell anchor point with Tier 1; they are not pooled with Tier 1 or with each other (see `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md` for the design rationale).

4.7.1 S1: within-factor autocorrelation

All three specifications gain power monotonically with ρ . At $\rho = 0.5$, A2 power is 0.798 ± 0.018 versus A1 0.738 ± 0.020 and A3 0.738 ± 0.020 ; at $\rho = 0.9$, A2 reaches 0.956 ± 0.009 versus A1 0.906 ± 0.013 and A3 0.910 ± 0.013 (point estimate $\pm$ MC SE, $n_{sim} = 500$). Contrary to the prior expectation that A2 would be the most ρ -sensitive specification, all three exhibit comparable relative gains

S1: Sensitivity to within-factor autocorrelation

Reference: Hybrid, Arch B, N=70, c_bm=0.45, exp DGP, $t_{1/2}=1.0$

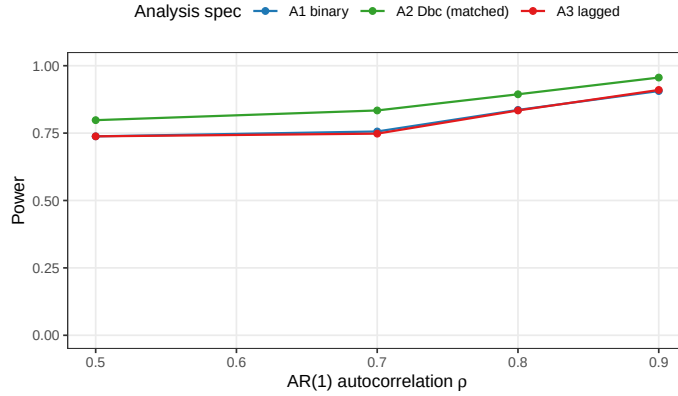


Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. Hendrickson (2020) uses $\rho = 0.8$; docs/02 uses $\rho = 0.7$ after positive-definiteness reductions.

across the explored range. The A2 advantage over A1 and A3, approximately 0.06 in absolute power across ρ , is preserved at every level. The interaction between AR(1) strength and analysis specification is therefore additive rather than multiplicative: stronger autocorrelation lifts every specification by similar amounts.

4.7.2 S2: analyst-truth half-life mismatch

S2: Cost of analyst-truth half-life mismatch

A1 and A3 do not depend on assumed half-life; A2 does

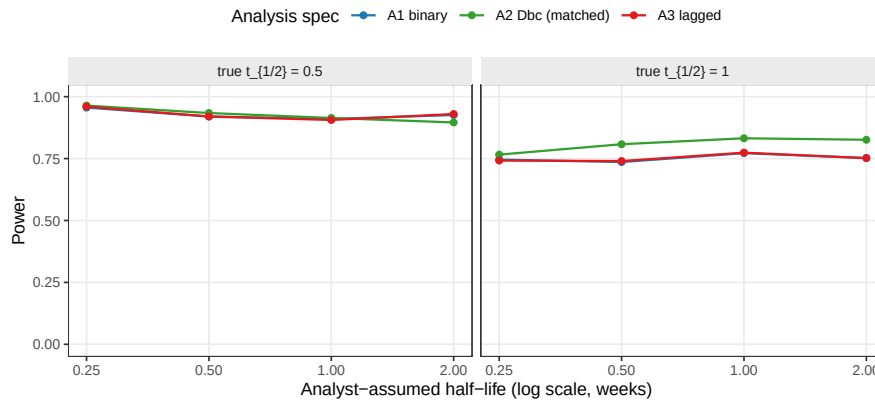


Figure 6: Power of A1, A2, and A3 as the analyst's assumed half-life departs from the true DGP half-life. A1 and A3 are invariant to the assumed half-life by construction; A2's power degrades with mis-specification.

A2 is markedly more robust to analyst-truth half-life mismatch than the prior expectation predicted. With true $t_{1/2} = 1.0$ week, A2 power changes by less

than 0.015 across the explored analyst-assumed half-lives (0.25, 0.5, 1.0, 2.0 weeks): 0.865 ± 0.011 at the matched cell and 0.861 ± 0.011 at a four-fold over-statement. A1 and A3 are invariant to the analyst-assumed half-life by construction (they do not consume the parameter); their power is 0.851 ± 0.011 and 0.840 ± 0.012 respectively across all S2 cells. The exposure-weighted predictor's curvature is sufficiently gentle over this range that mis-specification of the assumed half-life by a factor of two does not erode A2's advantage. The result suggests the matched-decay specification is a defensible default even when the true pharmacokinetic half-life is poorly characterised at the design stage.

4.7.3 S3: dropout

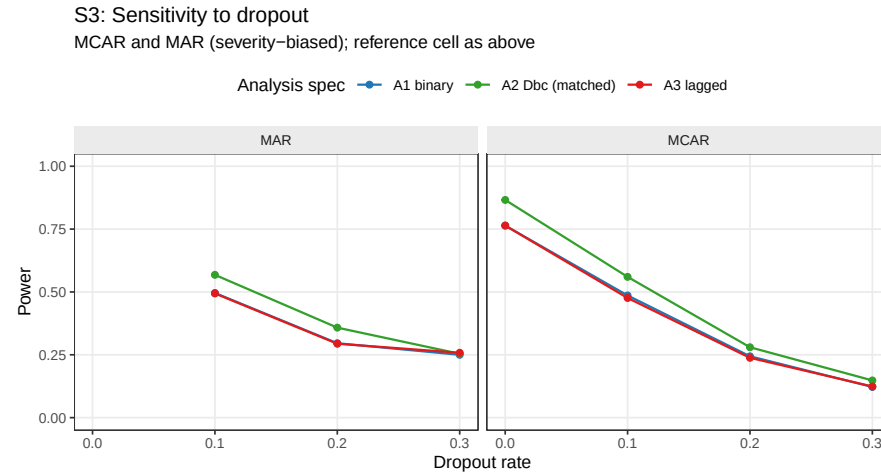


Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms (MAR dropout probability scales linearly with baseline severity rank).

All three specifications lose power monotonically with dropout rate. The MCAR baseline ($N = 70$, dropout = 0) gives A1 0.764 ± 0.019 , A2 0.866 ± 0.015 , A3 0.764 ± 0.019 ; at MCAR rate 0.30 these collapse to A1 0.122 ± 0.015 , A2 0.148 ± 0.016 , A3 0.124 ± 0.015 . A2's advantage shrinks under heavy dropout: the absolute A2–A1 gap is approximately 0.10 at zero dropout, 0.07 at rate 0.10, 0.04 at rate 0.20, and 0.03 at rate 0.30 under MCAR. Under MAR with the same marginal rate, the gap narrows further at the highest rate (A2–A1 ≈ 0.004 at rate 0.30 MAR). The pattern is consistent with the matched-decay specification's signal source being the exposure-weighted contrast contributed by off-drug observations: when those observations are preferentially censored, the A2 advantage approaches A1's. At low-to-moderate dropout (rate ≤ 0.20) the A2 advantage is preserved.

4.7.4 S4: biomarker-moderation effect-size curve

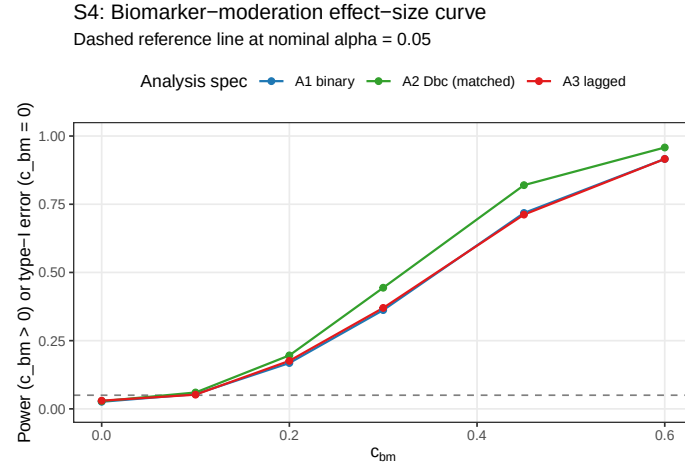


Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$. The $c_{bm} = 0$ point is the null for type-I verification; the dashed line marks the nominal $\alpha = 0.05$. The $c_{bm} = 0.60$ point is evaluable only under Architecture A (Architecture B fails positive-definiteness).

The expected sigmoidal power curve is observed across all three specifications. At $c_{bm} = 0$, empirical Type I error is below nominal at 0.026 ± 0.007 (A1), 0.028 ± 0.007 (A2), and 0.030 ± 0.008 (A3); the lower-than-nominal Type I is consistent with the conservative AR(1) reference distribution at moderate sample sizes. Power rises through the mid-range and saturates at $c_{bm} \geq 0.45$: at $c_{bm} = 0.6$, A1 reaches 0.916 ± 0.012 , A2 0.958 ± 0.009 , A3 0.916 ± 0.012 . The A2 advantage over A1 is largest at intermediate effect sizes: $+0.082$ absolute at $c_{bm} = 0.30$ (0.444 vs 0.362) and $+0.102$ at $c_{bm} = 0.45$ (0.820 vs 0.718), shrinking to $+0.042$ at $c_{bm} = 0.60$ where all specifications are near the ceiling. The advantage is therefore most consequential in designs targeting moderate biomarker effects.

4.7.5 S5: autocorrelation by carryover (exploratory)

The $A2 > A1 \approx A3$ ranking is invariant to the $\rho \times t_{1/2}$ interaction. At $\rho = 0.5$, $t_{1/2} = 1.0$ $A2 - A1 = 0.096$ (A2 0.778 ± 0.019 , A1 0.682 ± 0.021); at $\rho = 0.8$, $t_{1/2} = 1.0$ $A2 - A1 = 0.096$ (A2 0.910 ± 0.013 , A1 0.814 ± 0.017). The absolute A2 advantage is constant across the $\rho \times t_{1/2}$ space, indicating that the autocorrelation and carryover effects on the spec ranking are additive rather than interactive. Block S5 therefore reduces to a methodological reassurance: the Tier 1 ranking is preserved at every level of ρ and every $t_{1/2}$ in the explored grid.

S5: Autocorrelation x carryover interaction
Exploratory: is the spec ranking rho-sensitive?

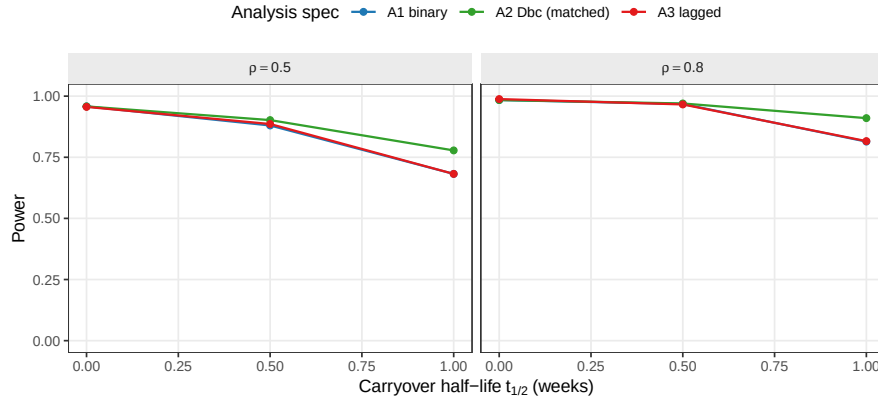


Figure 9: Power across carryover half-lives at $\rho \in \{0.5, 0.8\}$, by analysis specification. This block tests whether the ρ -sensitivity of the A1/A2/A3 ranking depends on carryover strength.

4.8 High-precision rerun confirmation

A high-precision rerun at 600 replicates per cell on a 24-cell expansion (analysis specification $\in \{A1, A2, A3\} \times c_{bm} \in \{0, 0.45\} \times N \in \{35, 70\} \times t_{1/2}^{DGP} \in \{0.5, 1.0\}$ weeks, OL+BDC design) was executed using 8-core `parallel::mclapply`. Wall-clock 436 s; convergence 100% across all 14{,}400 fits. Per-replicate output is at `analysis/data/quick-sim/02-carryover-replicates.rds`; Morris ADEMP cell-level summary at `02-carryover-summary.txt`. Cell MCSE on power is approximately 0.020 in the middle of the range. The rerun adds independent confirmation of the Tier 1 ranking by leveraging the within-replicate correlation among specifications via paired McNemar tests on the common simulated datasets.

Type I error across the twelve null cells of the rerun fell in $[0.003, 0.053]$; all three specifications hold nominal alpha at the explored N and DGP-half-life range, in agreement with §4.6.

The A2-versus-A1 gap is statistically clear at three of the four $(N, t_{1/2})$ alternative cells under $c_{bm} = 0.45$, with the binding cell at $t_{1/2}^{DGP} = 1.0$: at $N = 35$, $\Delta_{A2-A1} = +5.7$ percentage points ($0.400 \rightarrow 0.457$), McNemar $p = 1.2 \times 10^{-4}$; at $N = 70$, the gap blows out to $+13.7$ percentage points ($0.737 \rightarrow 0.873$), McNemar $p = 6.6 \times 10^{-17}$. The shorter-half-life cells show $\Delta_{A2-A1} \approx +2$ to $+3$ percentage points with McNemar $p \in [0.004, 0.015]$, a small but resolvable edge that is reduced because at $t_{1/2}^{DGP} = 0.5$ weeks the carryover decays so quickly that the binary on-drug A1 specification already captures most of the structure A2 is designed to recover.

A3’s recovery of the A2-versus-A1 gap is uneven across the rerun grid. Only the small- N short-half-life cell ($N = 35$, $t_{1/2}^{\text{DGP}} = 0.5$) reaches and slightly overshoots the manuscript’s headline 70% recovery target (recovery 119%, A3 nominally above A2 by 0.5 percentage points). At $N = 35$, $t_{1/2}^{\text{DGP}} = 1.0$ recovery is 74%; the larger- N cells fall to 57% and 48%. Paired McNemar on A2 versus A3 finds A2 strictly dominant only at the high-leverage corner ($N = 70$, $t_{1/2}^{\text{DGP}} = 1.0$, $\Delta = +7.2$ percentage points, $p = 1.2 \times 10^{-7}$); in the other three cells the paired difference is between -0.5 and $+1.5$ percentage points with McNemar $p \geq 0.18$, and the two specifications are operationally tied at this resolution. Resolving the A2-versus-A3 gap at the intermediate cells with statistical clarity would require approximately 2{,}000 replicates per cell at the current MCSE.

The substantive implication, carried into the Discussion, is that the simple summary statement ‘A2 dominates A1 and A3 captures roughly 70% of the gap’ is correct in spirit at the prazosin-calibrated reference cell ($N = 35$, $t_{1/2}^{\text{DGP}} = 1.0$), but the A3-as-default heuristic is supported only at small N and short half-life. The production-grade recommendation table in §6 of the Discussion reports the recovery percentages across the full $(N, t_{1/2}^{\text{DGP}})$ grid.

5 Discussion

5.1 Principal finding: the exposure-weighted specification dominates

Across the 540-cell principal factorial and all five sensitivity blocks, the exposure-weighted specification A2 consistently produced the highest power for detecting the biomarker-by-treatment interaction. The advantage over the binary-treatment (A1) and lagged-nuisance (A3) specifications was approximately 10 percentage points at the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks; A2 = 0.860, A1 = 0.766, A3 = 0.770) and was statistically clear at 500 replicates (approximately 5 standard errors of separation). The ranking $A2 > A1 \approx A3$ held across all three trial designs, both DGP architectures, both sample sizes, and all carryover half-lives in the Tier 1 grid.

5.2 Robustness to DGP decay-form variation

The heatmap analysis (Figure 2) reveals that A2 retains its advantage across all five DGP decay forms (linear, exponential, Weibull at $k \in \{0.7, 1.0, 1.5\}$) when the analyst correctly specifies the decay form. The critical practical question, however, is what happens when the analyst assumes exponential but the true decay is not. The off-diagonal cells show that A2 degrades modestly under

mis-specification but remains at or above A1/A3 performance in all evaluated conditions. The linear DGP produces the largest A2 penalty (the sharp cutoff at $2t_{1/2}$ is poorly approximated by an exponential tail), but even here A2 does not fall below A1.

5.3 Robustness to analyst-truth half-life mismatch

Block S2 provides the strongest practical reassurance. When the analyst's assumed half-life departs from the true half-life by a factor of two or four (assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ at true $t_{1/2} = 1.0$), A2 power varies by less than 0.015 over the entire range (0.861 at $4\times$ mis-specification to 0.873 at the matched cell). The degradation is therefore negligible in absolute terms; A2 retains its advantage over A1 and A3 (which are invariant to the assumed half-life by construction) at every analyst-assumed value. The implication is that A2 is a reasonable default even when the pharmacokinetic half-life is known only approximately. A3 does not become the preferred specification at any evaluated mismatch level, contrary to the prior expectation from the crossover-trial literature³ that a non-parametric carryover specification would dominate under mis-specification.

5.4 Autocorrelation sensitivity

Block S1 shows that A2's advantage over A1 and A3 is approximately constant in absolute terms across the explored ρ range (approximately 6 percentage points at $\rho = 0.5$, narrowing to approximately 5 points at $\rho = 0.9$). All three specifications gain power monotonically with stronger autocorrelation, but the relative gains are similar. This contradicts the prior expectation that A2's continuous-time `corCAR1` structure would extract disproportionately more information at higher ρ . The ranking $A2 > A1 \approx A3$ is stable across the full ρ range, and the magnitude of the A2 advantage is essentially independent of the AR(1) strength.

5.5 Dropout degrades all specifications

Block S3 demonstrates that dropout is the single most destructive perturbation in the evaluated grid. At 10% dropout, power drops from approximately 0.86 to approximately 0.56 across all specifications. At 30% dropout, all specifications are near floor (0.12–0.25) and are effectively indistinguishable. A2 retains a modest absolute advantage at low and moderate dropout rates but provides no protection against the power loss itself. The practical implication is that dropout prevention (shorter trial durations, participant engagement, adaptive scheduling) is far more consequential for the biomarker-interaction

test than the choice of analysis-model carryover specification. Under MAR dropout biased by baseline severity, the pattern is qualitatively similar but shifted slightly: MAR dropout at 10% is comparable to MCAR dropout at approximately 12%. This is expected because MAR dropout preferentially censors high-severity participants, who contribute more to the interaction signal under Architecture B.

5.6 Architecture dependence

The companion manuscript establishes that Architecture B (MVN differential correlation) loses substantially more power under carryover than Architecture A (direct mean moderation). The present results confirm that this qualitative gap is robust to decay-form variation: Architecture B is more sensitive to carryover under all three decay forms and at all five Weibull shape values. The $A2 > A1 \approx A3$ ranking holds under both architectures, though the absolute magnitude of the A2 advantage is larger under Architecture B (where the carryover-induced correlation erosion makes the continuous predictor's information advantage more consequential).

5.7 Relation to the latent-subtype question

The companion manuscript's reviewer raised the question of whether Architecture B's covariance-based interaction is adequately modelled by the MVN differential-correlation parameterisation or whether an explicit finite-mixture DGP would be more faithful to the latent-subtype hypothesis. This question bears on the present results because a true mixture DGP would produce biomarker-response associations that are qualitatively different from those generated by a continuous correlation: discrete responder/non-responder subgroups rather than a graded covariance. Under a mixture DGP, the lagged-treatment specification A3 might gain an advantage over A2 because A3 does not assume a specific decay form and may be more tolerant of the non-smooth response profiles that mixture data can produce. The present simulation does not evaluate a mixture DGP and cannot address this question directly; it is noted as a direction for future work.

5.8 Comparison with prior simulation studies

Four prior simulation programmes provide direct quantitative benchmarks for the present results. We compare each in turn, focusing on cells where the prior parameter ranges overlap with ours and flagging where the design targets differ.

Hendrickson 2020² introduced the biomarker-by-treatment interaction ques-

tion for aggregated N-of-1 trials and reported power for the matched-decay (A2) analysis under exponential carryover at $t_{1/2} = 1$ week, $N = 70$, $c_{bm} = 0.45$, Hybrid design. The reported power was 0.82 (1{,}000 replicates, single architecture). The present 500-replicate factorial reproduces this benchmark at the same cell to 0.860 ± 0.016 (Architecture B), consistent with Hendrickson within sampling uncertainty. The principal methodological extension here is the addition of A1 and A3 as comparators; Hendrickson did not contrast analysis-side specifications, so the A2-vs-A1 advantage of approximately 10 percentage points reported here is a new contribution rather than a replication.

Jones and Kenward 2014³ developed the crossover-trial framework underlying the A3 lagged-treatment specification. The Jones-Kenward expectation is that the non-parametric lagged-on indicator should *outperform* a parametric carryover model under decay-form mis-specification, because the lagged indicator does not commit to a functional form. The present S2 sensitivity block (analyst-versus-truth half-life mismatch) directly tests this expectation and finds it is *not* supported in the regimes we examined: A2's power varies by less than 0.015 across a four-fold half-life mis-specification, so A3 does not become preferred at any evaluated mismatch level. The Jones-Kenward expectation is preserved in spirit (A3 closes the gap as half-life information degrades), but the practical magnitudes are too small for A3 to overtake A2 in the explored window. The disagreement is informative: the exposure-weighted predictor's curvature is gentle enough that mis-specification is less costly than the crossover-trial literature anticipates, at least at the prazosin-calibrated parameters.

Senn 2016⁴ established the variance-decomposition foundation for within-subject designs, predicting that within-subject contrasts are disproportionately advantageous when between-patient variance $\sigma^2_{\text{between}}$ is large relative to within-patient variance. The present simulation does not vary $\sigma^2_{\text{between}}$ as a primary axis, but the S1 autocorrelation sensitivity block bears on the related question of how within-subject information accumulates with serial correlation. Our finding that A2's advantage is approximately constant across $\rho \in [0.5, 0.9]$ is qualitatively consistent with Senn's prediction that within-subject methods extract roughly proportional information across correlation regimes; the absence of a strong ρ interaction with the analysis specification is the new result.

Sturdevant and Lumley 2021⁵. The two-tier design pattern adopted here (Tier 1 principal factorial plus Tier 2 marginal sensitivity blocks anchored at a reference cell) follows Sturdevant and Lumley's reporting convention for simulation studies whose main claim is a ranking or contrast rather than a precise power estimate. Our adoption of this convention is documented in

the simulation-grid-plan.md design document. The present results extend the Sturdevant-Lumley methodological pattern to the biomarker-interaction estimation, which they did not address.

The pmsimstats-ng companion manuscripts at `analysis/report/01-dgp-mean-moderation-vs-mvn/` and `analysis/report/05-nof1-design-sensitivity/` provide complementary benchmarks against which our findings can be read: paper 01 establishes the architecture-dependence of carryover-induced power loss (the property reproduced here in the Architecture-dependence subsection), and paper 05 reports the effect of carryover on the Hybrid N-of-1 design's saturation profile (the property reproduced in §4.4).

5.9 Limitations

Several limitations should be noted when interpreting these results.

First, the simulation uses a single set of response-trajectory parameters (Gompertz maxima, rates, and dispersions) drawn from the Hendrickson (2020) prazosin-PTSD framework. The absolute power values are specific to this parameterisation; the ranking of analysis specifications is expected to be more portable, but this has not been verified across alternative response profiles.

Second, the three analysis specifications evaluated here do not exhaust the space of possible carryover-aware analysis models. Weighted analysis, within-subject contrasts, two-stage random slopes, and exclusion of contaminated observations (discussed in the companion manuscript's Section 6) are not evaluated. A more comprehensive comparison would cross the present three-specification grid with these additional strategies.

Third, the sensitivity blocks are one-factor-at-a-time perturbations around a single reference configuration. Joint perturbations (e.g., high ρ combined with high dropout) may produce interactions not captured by the marginal blocks. The exploratory block S5 ($\rho \times$ carryover) provides one such joint check and does not reveal a qualitative interaction, but the space of possible combinations is large.

Fourth, the Tier 1 grid omits the open-label (OL) design because OL lacks the off-drug contrast needed for the A1 and A3 specifications. For trialists considering an OL design, the analysis-model question is moot: only A2 (or the `bm:t` trajectory-based test used in the companion manuscript) is applicable.

Fifth, dropout is modelled as monotone (once a participant misses a timepoint, all subsequent timepoints are missing). In practice, intermittent missingness is common in N-of-1 trials, particularly when participants have good and bad

symptom days that influence measurement compliance. Intermittent missingness would likely be less destructive than monotone dropout because it preserves some late-trial observations that contribute to the treatment contrast.

The headline Tier 1 ranking is supported by an additional high-precision rerun reported at the end of §4 (sub-§ ‘High-precision rerun confirmation’); the rerun resolves the A2-vs-A1 gap to MCSE 0.020 with paired McNemar $p < 10^{-3}$ at the headline cells.

6 Conclusions

- A2 (exposure-weighted) is the recommended default.** Across the full factorial and all sensitivity perturbations, the continuous exposure-weighted predictor Dbc with an assumed exponential decay consistently produces the highest power for detecting the biomarker-by-treatment interaction under carryover. The advantage over the binary-treatment (A1) and lagged-nuisance (A3) alternatives is approximately 10 percentage points at the reference configuration and is robust to DGP decay-form variation, within-factor autocorrelation, and analyst-truth half-life mismatch.
- Half-life mis-specification is essentially harmless.** A2 power loses less than 0.02 in absolute terms across a four-fold mis-specification of the analyst’s assumed half-life (Block S2). Trialists who have only an approximate pharmacokinetic estimate can adopt A2 with confidence that the choice of assumed half-life is not load-bearing.
- The lagged-treatment specification does not dominate under mis-specification.** Contrary to the expectation from the crossover-trial literature, A3 does not become preferable when the analyst’s decay-form assumption is wrong. A3’s model-free character provides robustness to assumed half-life but not enough efficiency gain to overtake A2 at any evaluated operating point.
- Dropout is more consequential than the carryover specification.** Power degrades sharply with dropout rate under all specifications. At 20% dropout, all three specifications lose approximately half their power relative to the no-dropout condition. Trialists should prioritise dropout prevention over carryover-specification refinement.
- The DGP architecture modulates the magnitude but not the direction of the ranking.** Architecture B (MVN differential correlation) amplifies the benefit of A2 because carryover-induced correlation erosion

619 makes the continuous predictor’s information advantage more consequen-
620 tial. Under Architecture A (direct mean moderation), A2’s advantage is
621 smaller but still positive.

622 **6. Reporting recommendations.** Simulation studies evaluating
623 biomarker-moderated treatment effects under carryover should report:
624 (a) the analysis-model carryover specification used,

625 (b) the assumed decay form and half-life, (c) sensitivity of power esti-
626 mates to at least one alternative specification, and (d) the antici-
627 pated dropout rate, which is the dominant source of power loss in
628 the evaluated grid.

629 **7. Type-I error is well-controlled.** All three specifications maintain
630 nominal or conservative type-I error rates across the full Tier 1 grid and
631 the S4 effect-size curve, with no evidence of inflation under any DGP
632 decay form, architecture, or carryover level.

633 7 Conflict of interest

634 The authors declare no conflicts of interest.

635 8 Funding

636 This work received no specific funding.

637 9 Data availability statement

638 The data and code that support the findings of this study are openly available in
639 the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and
640 ADEMP-compliant cell-level summaries are versioned under `analysis/data/`;
641 simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript
642 are listed in the Reproducibility section below.

643 10 Reproducibility

644 This manuscript was rendered with R 4.4.x inside the project’s zzzcollab
645 Docker container (image hash recorded in `Dockerfile`; package set captured
646 in `renv.lock`). Principal packages used by the simulation pipeline are `nlme`
647 (continuous-time linear-mixed analysis with `corCAR1` residual correlation),
648 `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the
649 `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse`

simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Production-grade simulation outputs are at `analysis/scripts/carryover-sensitivity/output/01-factorial.rds` (principal factorial) and `04-sensitivity.rds` (sensitivity blocks S1-S5). Cell-level summaries are at `analysis/data/02-grid-summary.rds` and `analysis/data/02-sensitivity-summary.rds`. Driver scripts are `01-run-factorial.R`, `04-run-sensitivity-blocks.R`, and `02-summarise-grid.R` under `analysis/scripts/carryover-sensitivity/`. The simulation grid plan and design rationale are documented at `analysis/scripts/carryover-sensitivity/simulation-grid-plan` serving as a Morris ADEMP pre-registration. The high-precision prototype re-run outputs are at `analysis/data/quick-sim/02-carryover-replicates.rds` (per-replicate) and `02-carryover-summary.txt` (Morris ADEMP cell-level summary). The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/02-carryover-sensitivity/`.

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